PYTHON ACTIVITY-BASED LEARNING

A Collection of 10 Python Case Studies

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Case Study 1: Progressive Electricity Billing

1. Problem Statement

Calculate an electricity bill using progressive unit rates.

2. Objectives

• Apply different rates to unit slabs.

• Practise variables, arithmetic and conditions.

3. Assumptions

• First 100 units: ■2 each; next 100: ■3 each; above 200: ■5 each.

• No fixed charge or tax; negative units are invalid.

4. Algorithm

• Read units.

• Reject negative units.

• For up to 100, charge units × 2.

• For up to 200, charge 200 + (units – 100) × 3.

• Above 200, charge 500 + (units – 200) × 5.

• Display bill.

5. Flowchart

Vertical flowchart: See page 14.

6. Pseudocode

```
INPUT units; IF units<0: invalid; ELIF units<=100: bill=units*2; ELIF units<=200: bill=200+(units-100)*3; ELSE: bill=500+(units-200)*5;
```

7. Python Source Code

```
units = int(input("Enter units: "))
if units < 0:
    print("Invalid units")
elif units <= 100:
    bill = units * 2
elif units <= 200:
    bill = 200 + (units - 100) * 3
else:
    bill = 500 + (units - 200) * 5
if units >= 0:
    print("Electricity bill: Rs.", bill)
```

8. Test Cases

Test Input	Expected Result
50	■100
100	■200
150	■350
250	■750
-5	Invalid units

9. Sample Program Output

```
Enter units: 250
Electricity bill: Rs. 750
```

10. Conclusion

Progressive billing is calculated using arithmetic and if-elif-else statements.

Case Study 2: Employee Payroll and Deduction Analysis

1. Problem Statement

Calculate an employee's gross and net salary from basic salary, allowances and deductions.

2. Objectives

• Calculate percentage-based salary components.

• Practise arithmetic and conditions.

3. Assumptions

• HRA = 20% of basic; DA = 10% of basic; deduction = 5% of gross.

• Salary cannot be negative.

4. Algorithm

• Read employee name and basic salary.

• Reject negative salary.

• Calculate HRA and DA.

• Calculate gross salary.

• Calculate 5% deduction and net salary.

• Display payslip.

5. Flowchart

Vertical flowchart: See page 15.

6. Pseudocode

```
INPUT name,basic; IF basic<0: invalid; ELSE hra=basic*0.20; da=basic*0.10; gross=basic+hra+da; deduction=gross*0.05; net=gross-deductio
```

7. Python Source Code

```
name = input("Employee name: ")
basic = float(input("Basic salary: "))
if basic < 0:
    print("Invalid salary")
else:
    hra = basic * 0.20
    da = basic * 0.10
    gross = basic + hra + da
    deduction = gross * 0.05
    net = gross - deduction
    print("Employee:", name)
    print("Gross salary: Rs.", round(gross, 2))
    print("Deduction: Rs.", round(deduction, 2))
    print("Net salary: Rs.", round(net, 2))
```

8. Test Cases

Test Input	Expected Result
■20,000	Gross ■26,000; deduction ■1,300; net ■24,700
■30,000	Gross ■39,000; deduction ■1,950; net ■37,050
-500	Invalid salary

9. Sample Program Output

```
Employee name: Ravi
Basic salary: 20000
Gross salary: Rs. 26000.0
Deduction: Rs. 1300.0
Net salary: Rs. 24700.0
```

10. Conclusion

The payroll program calculates allowances, deductions and net salary using arithmetic and conditions.

Case Study 3: Retail Billing with Multiple Discount Rules

1. Problem Statement

Read item prices, calculate the total and apply a discount based on the purchase amount.

2. Objectives

• Store prices in a list.

• Use a loop to total prices and conditions to select a discount.

3. Assumptions

• Total ≥ ■5000: 10% discount; ■2000–■4999.99: 5%; below ■2000: no discount.

• Prices must not be negative.

4. Algorithm

• Read item count and create an empty list.

• Input each price and append it.

• Calculate subtotal using sum().

• Choose discount rate using conditions.

• Calculate and display final amount.

5. Flowchart

Vertical flowchart: See page 16.

6. Pseudocode

```
INPUT count; prices=[]; REPEAT count times: INPUT price; APPEND price; subtotal=SUM(prices); IF subtotal>=5000 rate=.10; ELIF subtotal>
```

7. Python Source Code

```
count = int(input("Number of items: "))
prices = []
for i in range(count):
    price = float(input("Price of item " + str(i + 1) + ": "))
    if price < 0:
        print("Invalid price")
        break
    prices.append(price)
else:
    subtotal = sum(prices)
    if subtotal >= 5000:
        rate = 0.10
    elif subtotal >= 2000:
        rate = 0.05
    else:
        rate = 0
    discount = subtotal * rate
    print("Subtotal: Rs.", subtotal)
    print("Discount: Rs.", discount)
    print("Final amount: Rs.", subtotal - discount)
```

8. Test Cases

Test Input	Expected Result
■1000 + ■1200	Subtotal ■2200; discount ■110; final ■2090
■3000 + ■2500	Subtotal ■5500; discount ■550; final ■4950
■500 + ■700	Subtotal ■1200; no discount

9. Sample Program Output

```
Number of items: 2
Price of item 1: 3000
Price of item 2: 2500
Subtotal: Rs. 5500.0
Discount: Rs. 550.0
Final amount: Rs. 4950.0
```

10. Conclusion

The billing program uses a list, loop and conditions to calculate discounts.

Case Study 4: Attendance Recovery Computation

1. Problem Statement

Calculate attendance percentage and find the additional consecutive classes needed to reach 75%.

2. Objectives

• Calculate attendance percentage.

• Use a while loop to find required additional classes.

3. Assumptions

• Required attendance is 75%.

• Conducted classes must be positive and attended cannot exceed conducted.

4. Algorithm

• Input conducted and attended classes.

• Validate values.

• Calculate percentage.

• While percentage is below 75, add one attended and one conducted class.

• Display current percentage and extra classes.

5. Flowchart

Vertical flowchart: See page 17.

6. Pseudocode

```
INPUT conducted,attended; validate; percentage=attended/conducted*100; extra=0; WHILE percentage<75: conducted+=1; attended+=1; extra+=
```

7. Python Source Code

```
conducted = int(input("Total classes conducted: "))
attended = int(input("Classes attended: "))
if conducted <= 0 or attended < 0 or attended > conducted:
    print("Invalid attendance details")
else:
    original = attended / conducted * 100
    extra = 0
    while attended / conducted * 100 < 75:
        attended += 1
        conducted += 1
        extra += 1
    print("Current attendance:", round(original, 2), "%")
    print("Additional classes required:", extra)
    print("Attendance after recovery:", round(attended / conducted * 100, 2), "%")
```

8. Test Cases

Test Input	Expected Result
100 conducted, 60 attended	60%; 60 additional classes
80 conducted, 64 attended	80%; 0 additional classes
50 conducted, 30 attended	60%; 30 additional classes

9. Sample Program Output

```
Total classes conducted: 100
Classes attended: 60
Current attendance: 60.0 %
Additional classes required: 60
Attendance after recovery: 75.0 %
```

10. Conclusion

The attendance program uses a while loop and conditions to calculate recovery classes.

Case Study 5: ATM Transaction Sequence Simulation

1. Problem Statement

Simulate deposits, withdrawals, balance enquiry and exit; record valid transactions in a list.

2. Objectives

• Use a list to store transaction history.

• Practise loops, conditions and balance updates.

3. Assumptions

• Starting balance is ■5000.

• Amounts must be positive; withdrawal cannot exceed balance.

4. Algorithm

• Set balance to 5000 and transactions to an empty list.

• Display menu repeatedly.

• Deposit: add amount and record it.

• Withdraw: validate amount and balance, then subtract and record.

• Balance: display balance; Exit: stop.

• Display history and final balance.

5. Flowchart

Vertical flowchart: See page 18.

6. Pseudocode

```
balance=5000; transactions=[]; WHILE true: INPUT choice; IF deposit validate/add; ELIF withdraw validate/subtract; ELIF balance print;
```

7. Python Source Code

```
balance = 5000.0
transactions = []
while True:
    print("\n1.Deposit 2.Withdraw 3.Balance 4.Exit")
    choice = input("Choice: ")
    if choice == "1":
        amount = float(input("Deposit amount: "))
        if amount > 0:
            balance += amount
            transactions.append("Deposit: " + str(amount))
        else:
            print("Enter a positive amount")
    elif choice == "2":
        amount = float(input("Withdrawal amount: "))
        if amount <= 0:
            print("Enter a positive amount")
        elif amount > balance:
            print("Insufficient balance")
        else:
            balance -= amount
            transactions.append("Withdrawal: " + str(amount))
    elif choice == "3":
        print("Balance: Rs.", balance)
    elif choice == "4":
        break
    else:
        print("Invalid choice")
print("Transactions:", transactions)
print("Final balance: Rs.", balance)
```

8. Test Cases

Test Input	Expected Result
Deposit ■1000	Balance becomes ■6000
Withdraw ■1500	Balance becomes ■3500
Withdraw ■7000	Insufficient balance
Choose 4	Session ends

9. Sample Program Output

```
Choice: 1
Deposit amount: 1000
Choice: 3
Balance: Rs. 6000.0
Choice: 4
Transactions: ['Deposit: 1000.0']
Final balance: Rs. 6000.0
```

10. Conclusion

The ATM simulation uses a menu loop, conditions and a list to maintain transaction history.

Case Study 6: Password Policy Validation Engine

1. Problem Statement

Check whether a password contains at least eight characters, an uppercase letter, a lowercase letter and a digit.

2. Objectives

• Practise string operations and character checks.

• Use a loop and Boolean variables.

3. Assumptions

• Minimum length is 8.

• At least one uppercase, lowercase and digit are required.

• Special characters are optional.

4. Algorithm

• Input password.

• Set uppercase, lowercase and digit flags to False.

• Check every character and update flags.

• Check length and all flags.

• Display valid or invalid.

5. Flowchart

Vertical flowchart: See page 19.

6. Pseudocode

```
INPUT password; upper=lower=digit=false; FOR each character: check isupper/islower/isdigit; IF length>=8 AND all flags: valid ELSE inva
```

7. Python Source Code

```
password = input("Enter password: ")
has_upper = has_lower = has_digit = False
for ch in password:
    if ch.isupper():
        has_upper = True
    elif ch.islower():
        has_lower = True
    elif ch.isdigit():
        has_digit = True
if len(password) >= 8 and has_upper and has_lower and has_digit:
    print("Valid password")
else:
    print("Invalid password")
    print("Use 8+ characters with uppercase, lowercase and a digit.")
```

8. Test Cases

Test Input	Expected Result
Hello123	Valid password
hello123	Invalid: no uppercase
HELLO123	Invalid: no lowercase
HelloWorld	Invalid: no digit
Hi1	Invalid: too short

9. Sample Program Output

```
Enter password: Hello123
Valid password
```

10. Conclusion

The validator uses strings, loops and conditions to check password requirements.

Case Study 7: Text Compression Using Character Runs

1. Problem Statement

Compress consecutive repeated characters by writing each character followed by its repetition count, such as AAABB becoming A3B2.

2. Objectives

• Practise string indexing and traversal.

• Build a result using concatenation and a loop.

3. Assumptions

• Input is case-sensitive and contains no spaces.

• Every run includes its count, even when the count is 1.

4. Algorithm

• Input text and create an empty result.

• If not empty, set count to 1.

• Compare each character with the previous one.

• Increase count for a match; otherwise append previous character and count.

• Append the final run and display.

5. Flowchart

Vertical flowchart: See page 20.

6. Pseudocode

```
INPUT text; result=''; IF text not empty: count=1; FOR i from 1: IF text[i]==text[i-1] count++; ELSE append previous+count and reset; a
```

7. Python Source Code

```
text = input("Enter text (no spaces): ")
compressed = ""
if text:
    count = 1
    for i in range(1, len(text)):
        if text[i] == text[i - 1]:
            count += 1
        else:
            compressed += text[i - 1] + str(count)
            count = 1
    compressed += text[-1] + str(count)
print("Compressed text:", compressed)
```

8. Test Cases

Test Input	Expected Result
AAABB	A3B2
WWWWAA	W4A2
ABC	A1B1C1
A	A1
Empty input	Empty result

9. Sample Program Output

```
Enter text (no spaces): AAABB
Compressed text: A3B2
```

10. Conclusion

The program demonstrates string processing, loops, indexing and concatenation.

Case Study 8: Nested Loop Calendar Generator

1. Problem Statement

Display a simple month calendar from the number of days and the weekday on which the month starts.

2. Objectives

• Practise loops and pattern generation.

• Use conditions to print calendar rows.

3. Assumptions

• Days must be 28–31.

• Start weekday is 0=Sunday through 6=Saturday.

• No month/year calculation is performed.

4. Algorithm

• Input days and starting weekday.

• Validate both values.

• Print weekday headings and leading blank cells.

• Loop through dates and print each in a fixed-width column.

• Start a new row after Saturday.

5. Flowchart

Vertical flowchart: See page 21.

6. Pseudocode

```
INPUT days,start; validate; PRINT headings; print start blank cells; FOR date=1..days: print date; IF (start+date) MOD 7=0: newline; ST
```

7. Python Source Code

```
days = int(input("Days in month (28 to 31): "))
start = int(input("Starting weekday (0=Sun, 6=Sat): "))
if days < 28 or days > 31 or start < 0 or start > 6:
    print("Invalid calendar details")
else:
    print("Sun Mon Tue Wed Thu Fri Sat")
    for blank in range(start):
        print("    ", end="")
    for date in range(1, days + 1):
        print(f"{date:3}", end=" ")
        if (start + date) % 7 == 0:
            print()
    print()
```

8. Test Cases

Test Input	Expected Result
30 days, start 0	Dates begin under Sunday
31 days, start 3	Dates begin under Wednesday
27 days	Invalid calendar details
30 days, start 7	Invalid calendar details

9. Sample Program Output

```
Days in month (28 to 31): 30
Starting weekday (0=Sun, 6=Sat): 0
Sun Mon Tue Wed Thu Fri Sat
  1   2   3   4   5   6   7
  8   9  10  11  12  13  14
 15  16  17  18  19  20  21
 22  23  24  25  26  27  28
 29  30
```

10. Conclusion

The calendar program generates a table-like pattern using loops, conditions and formatted output.

Case Study 9: Student Marks Matrix Analysis

1. Problem Statement

Store marks of students in a nested list and calculate student totals, averages and subject averages.

2. Objectives

• Represent tabular data using nested lists.

• Use nested loops to process rows and columns.

3. Assumptions

• Three students and three subjects are used.

• Marks are between 0 and 100.

• Rows represent students; columns represent subjects.

4. Algorithm

• Create a nested list of marks.

• For each student row, calculate total and average.

• For each subject column, add marks across students.

• Calculate and display subject averages.

5. Flowchart

Vertical flowchart: See page 22.

6. Pseudocode

```
marks=[[80,75,90],[65,85,70],[90,95,88]]; FOR each row: total=SUM(row), average=total/3; FOR each column: sum values across rows and di
```

7. Python Source Code

```
marks = [[80, 75, 90], [65, 85, 70], [90, 95, 88]]
names = ["Anu", "Balu", "Chandu"]
subjects = ["Python", "Maths", "English"]

for i in range(len(marks)):
    total = sum(marks[i])
    average = total / len(marks[i])
    print(names[i], "Total:", total, "Average:", round(average, 2))

print("Subject averages:")
for col in range(len(subjects)):
    total = 0
    for row in range(len(marks)):
        total += marks[row][col]
    print(subjects[col], ":", round(total / len(marks), 2))
```

8. Test Cases

Test Input	Expected Result
Anu: 80,75,90	Total 245; average 81.67
Balu: 65,85,70	Total 220; average 73.33
Chandu: 90,95,88	Total 273; average 91.00
Python marks 80,65,90	Subject average 78.33

9. Sample Program Output

```
Anu Total: 245 Average: 81.67
Balu Total: 220 Average: 73.33
Chandu Total: 273 Average: 91.0
Subject averages:
Python : 78.33
Maths : 85.0
English : 82.67
```

10. Conclusion

The nested-list program processes student rows and subject columns to calculate useful statistics.

Case Study 10: Tuple-Based Employee Shift Conflict Detection

1. Problem Statement

Store employee shifts as tuples and determine whether two shifts overlap.

2. Objectives

• Create and access tuples.

• Use tuple unpacking.

• Compare time ranges using conditions.

3. Assumptions

• Shift format is (employee_name, start_hour, end_hour).

• Hours are whole numbers from 0 to 23; overnight shifts are excluded.

• Two shifts overlap when start1 < end2 and start2 < end1.

4. Algorithm

• Create two shift tuples.

• Unpack each tuple into name, start and end.

• Validate time ranges.

• Apply the overlap condition.

• Display conflict or no conflict.

5. Flowchart

Vertical flowchart: See page 23.

6. Pseudocode

```
shift1=(name1,start1,end1); shift2=(name2,start2,end2); UNPACK; validate; IF start1<end2 AND start2<end1: conflict ELSE no conflict.
```

7. Python Source Code

```
shift1 = ("Anu", 9, 13)
shift2 = ("Balu", 12, 16)
name1, start1, end1 = shift1
name2, start2, end2 = shift2

if not (0 <= start1 < end1 <= 23 and 0 <= start2 < end2 <= 23):
    print("Invalid shift timing")
elif start1 < end2 and start2 < end1:
    print("Shift conflict detected")
    print(name1, "and", name2, "have overlapping shifts.")
else:
    print("No shift conflict")
```

8. Test Cases

Test Input	Expected Result
Anu 9–13; Balu 12–16	Conflict detected
Anu 9–13; Balu 13–17	No conflict
Anu 8–12; Balu 14–18	No conflict
Start 15, end 10	Invalid shift timing

9. Sample Program Output

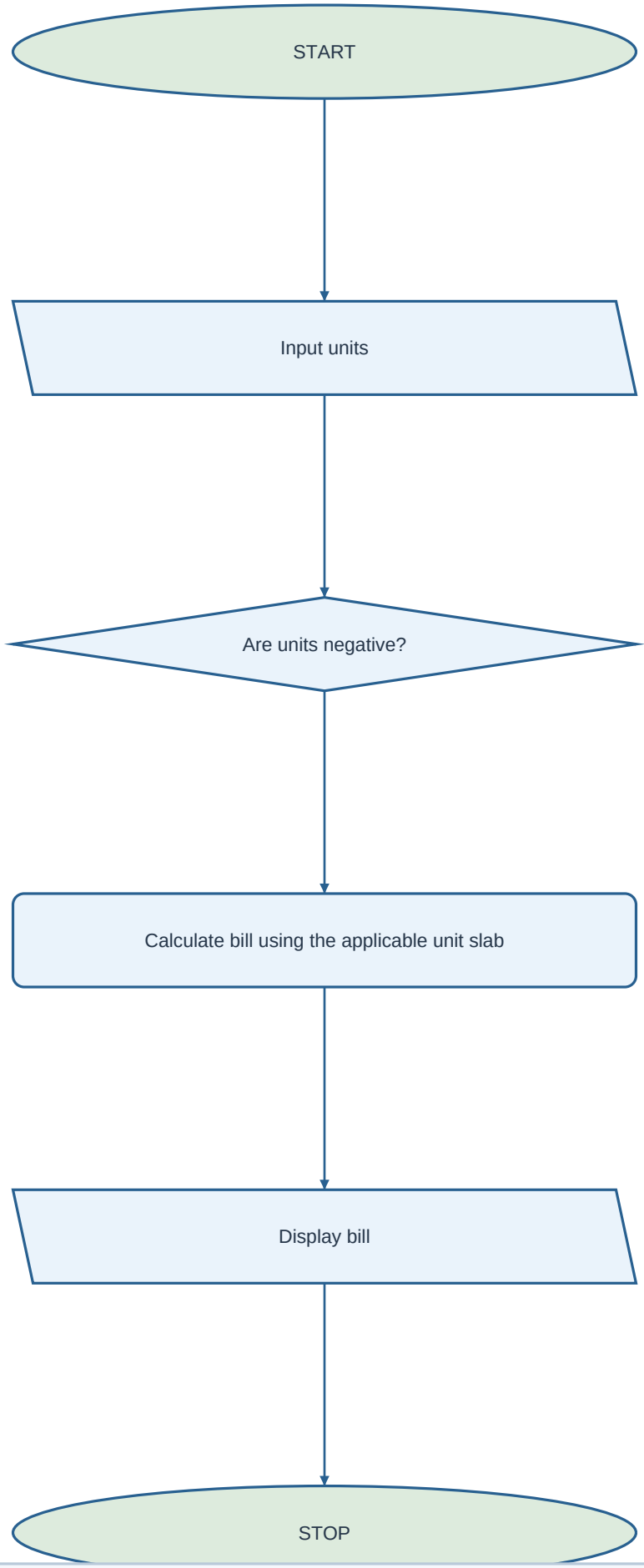
```
Shift conflict detected
Anu and Balu have overlapping shifts.
```

10. Conclusion

The program uses tuples and tuple unpacking to compare employee shift times.

Case Study 1: Progressive Electricity Billing

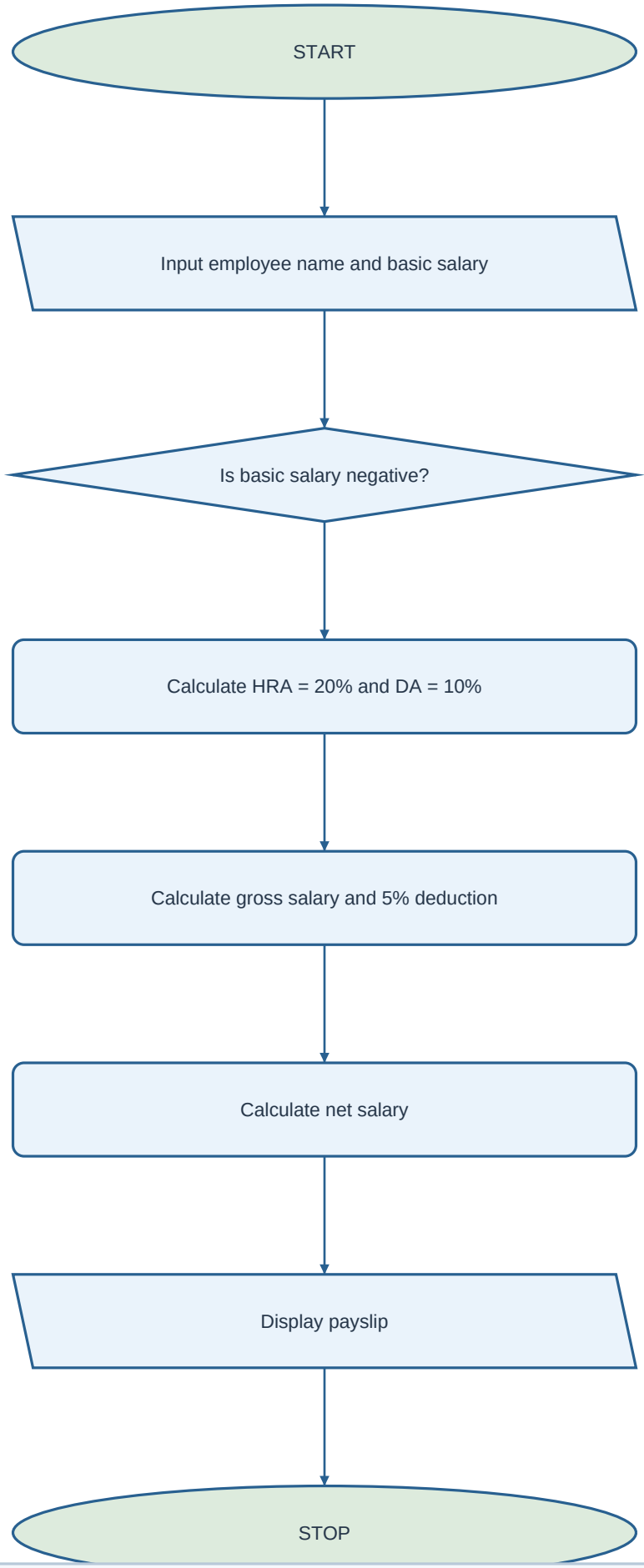
VERTICAL FLOWCHART



Decision paths: NO: Calculate bill | YES: Display “Invalid units”

Case Study 2: Employee Payroll and Deduction Analysis

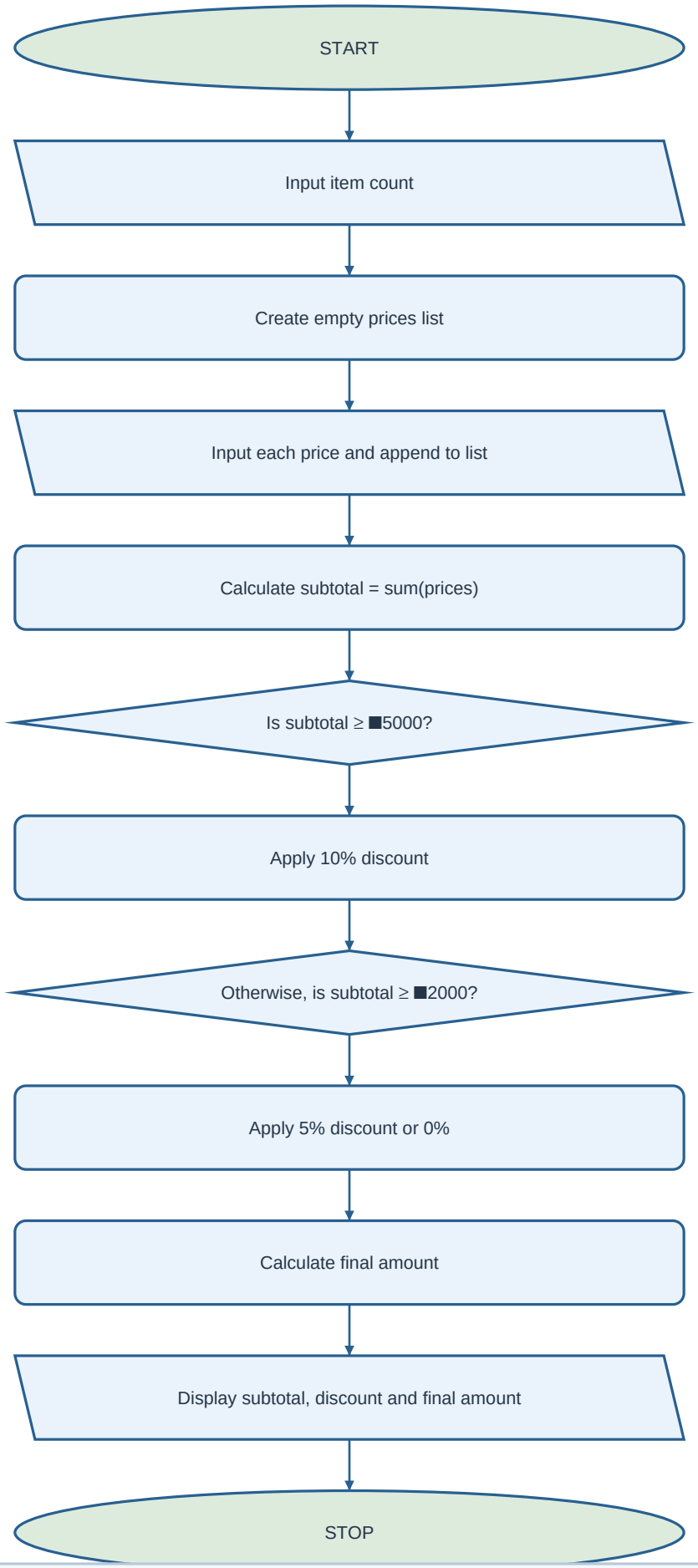
VERTICAL FLOWCHART



Decision paths: NO: Calculate HRA and DA | YES: Display “Invalid salary”

Case Study 3: Retail Billing with Multiple Discount Rules

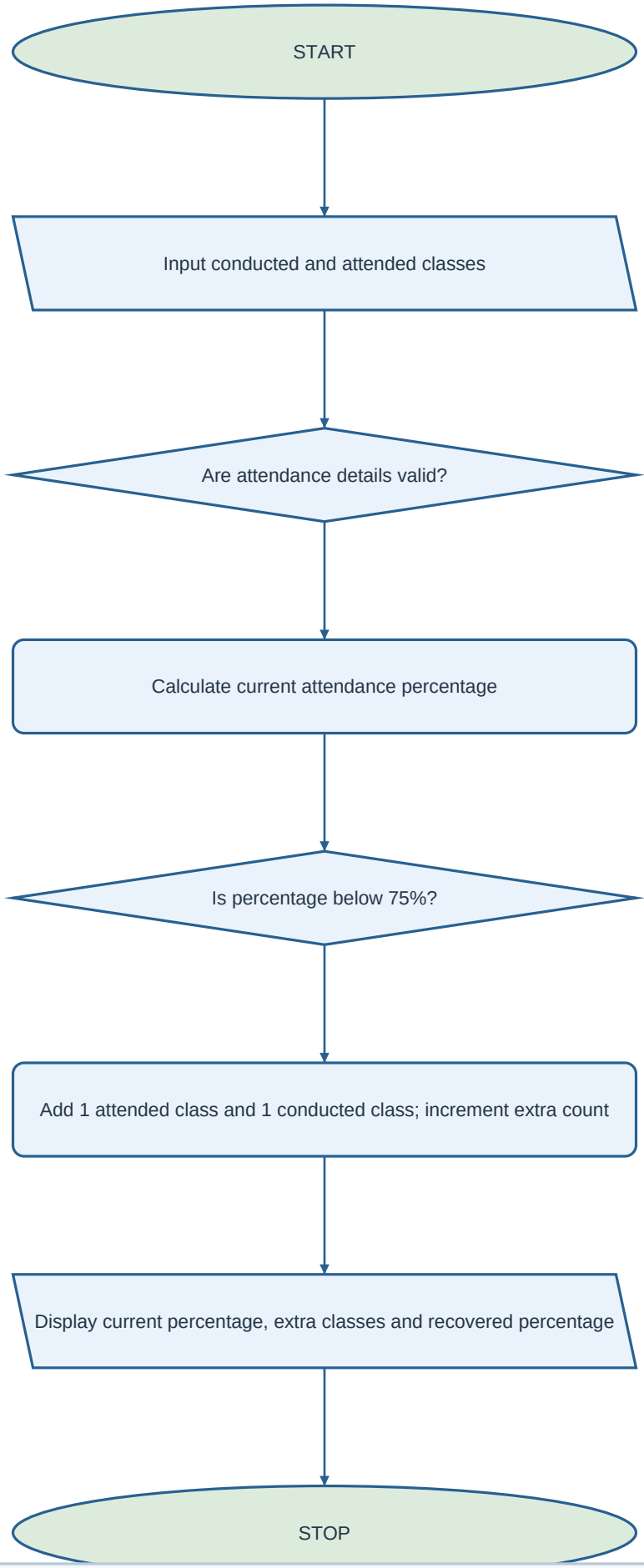
VERTICAL FLOWCHART



Decision paths: YES: Apply 10% | NO: Check next slab

Case Study 4: Attendance Recovery Computation

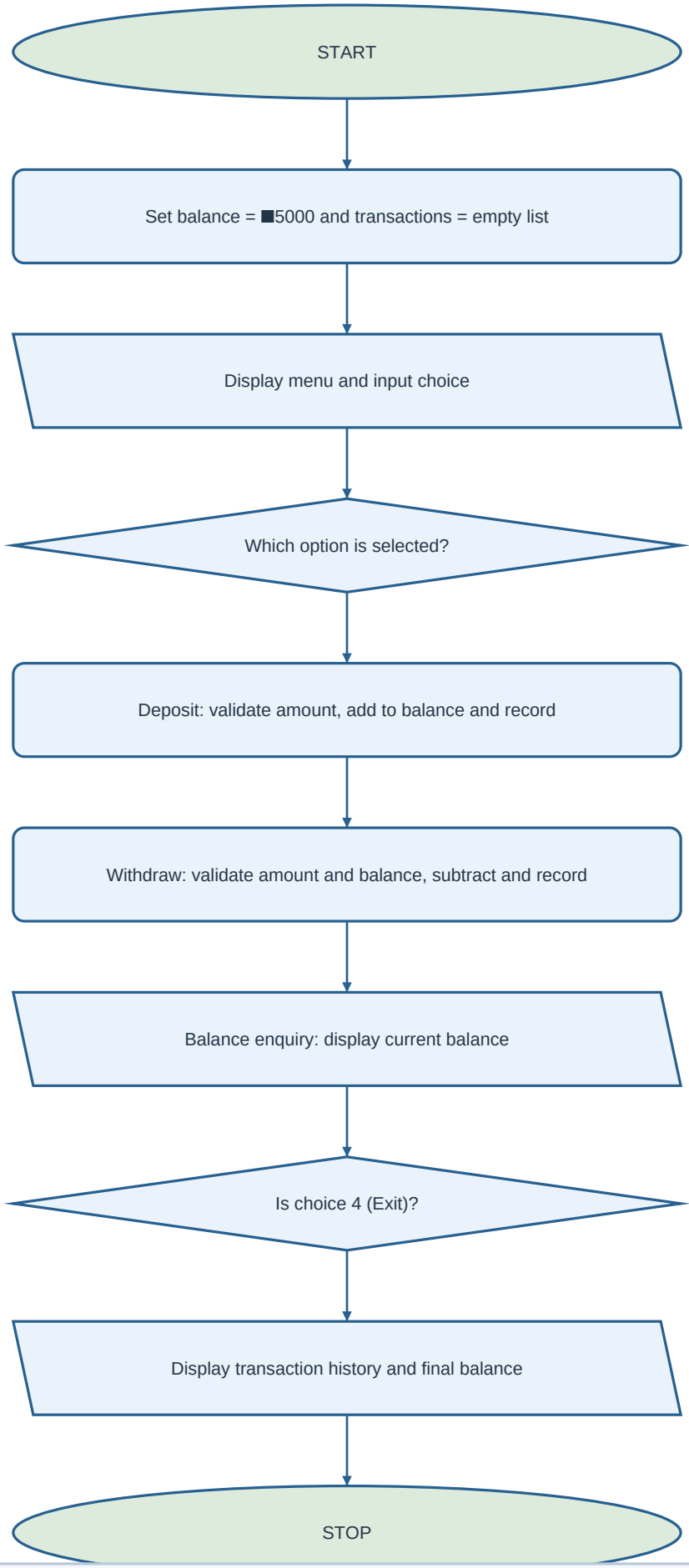
VERTICAL FLOWCHART



Decision paths: NO: Display invalid attendance details | YES: Continue calculation | YES: Repeat until percentage reaches 75% | NO: Display results

Case Study 5: ATM Transaction Sequence Simulation

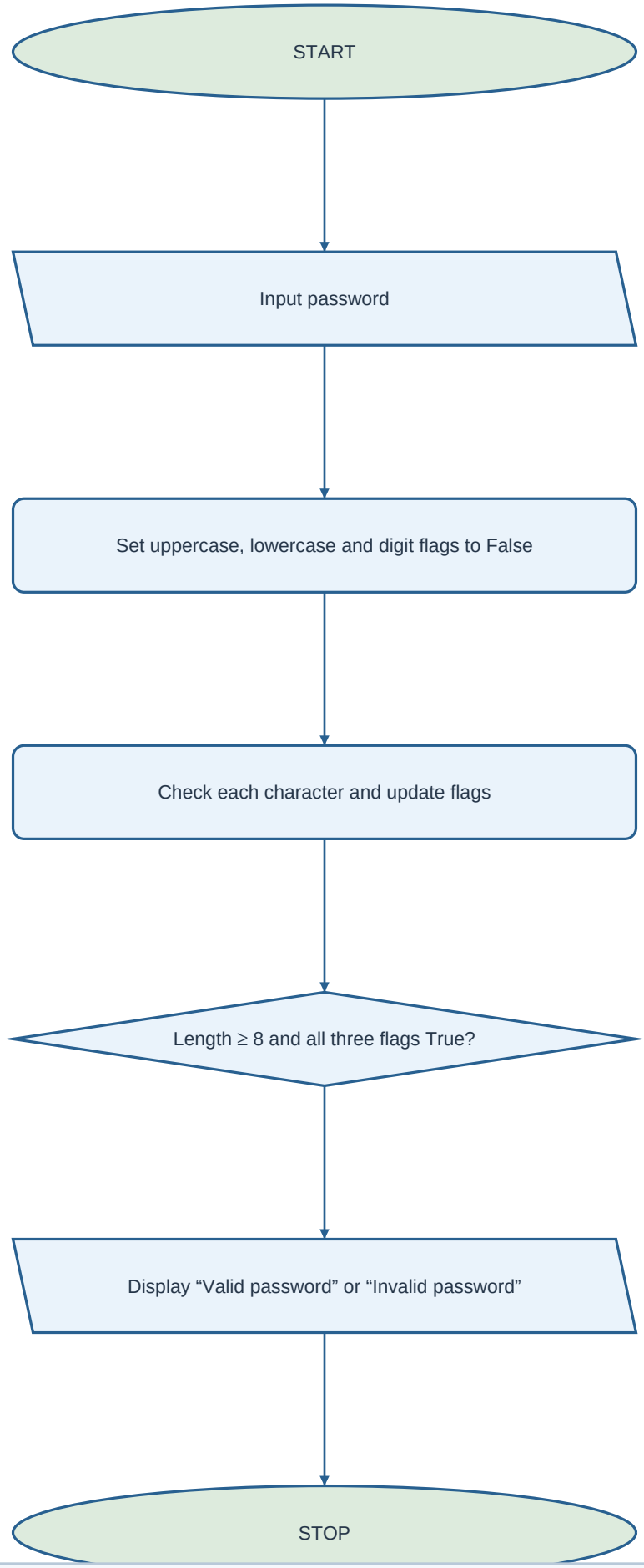
VERTICAL FLOWCHART



Decision paths: 1: Deposit | 2: Withdraw | 3: Balance | 4: Exit | NO: Return to menu

Case Study 6: Password Policy Validation Engine

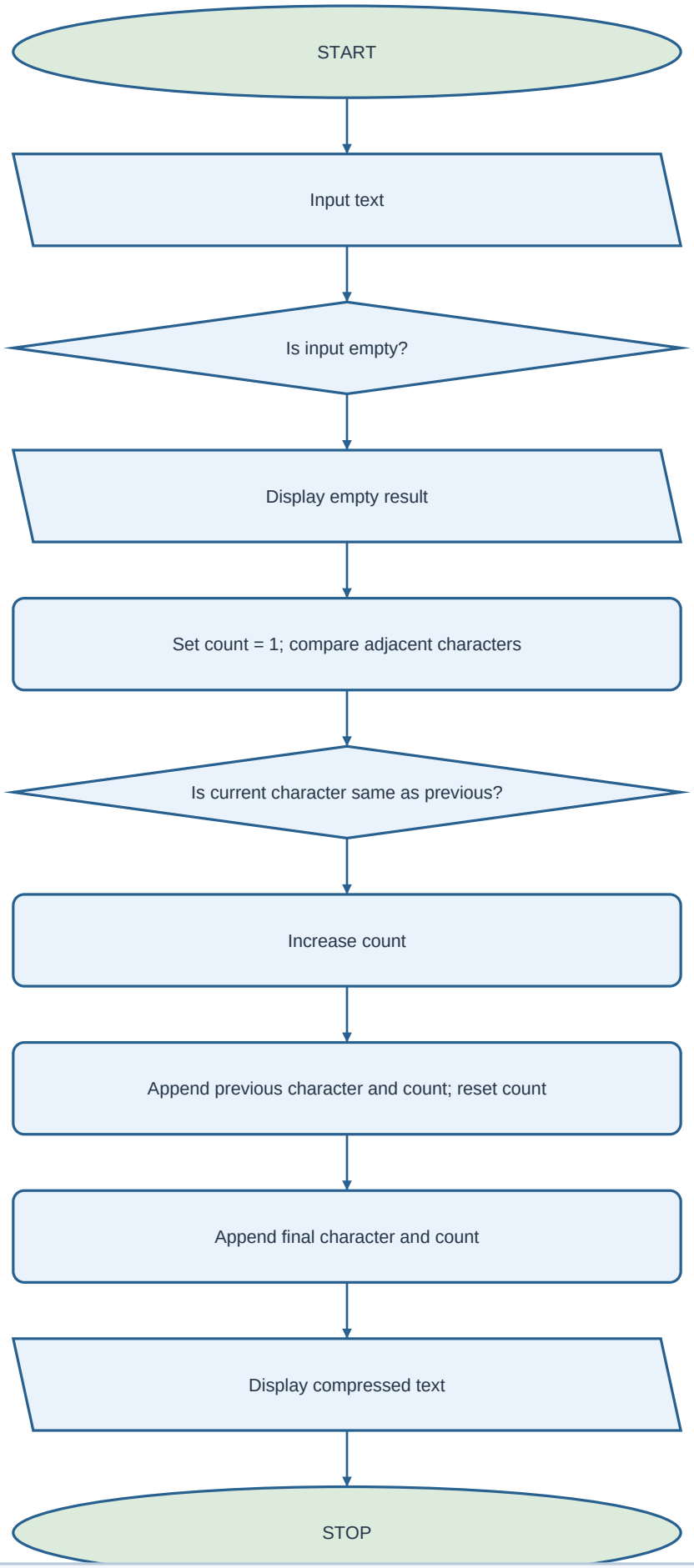
VERTICAL FLOWCHART



Decision paths: YES: Valid password | NO: Invalid password

Case Study 7: Text Compression Using Character Runs

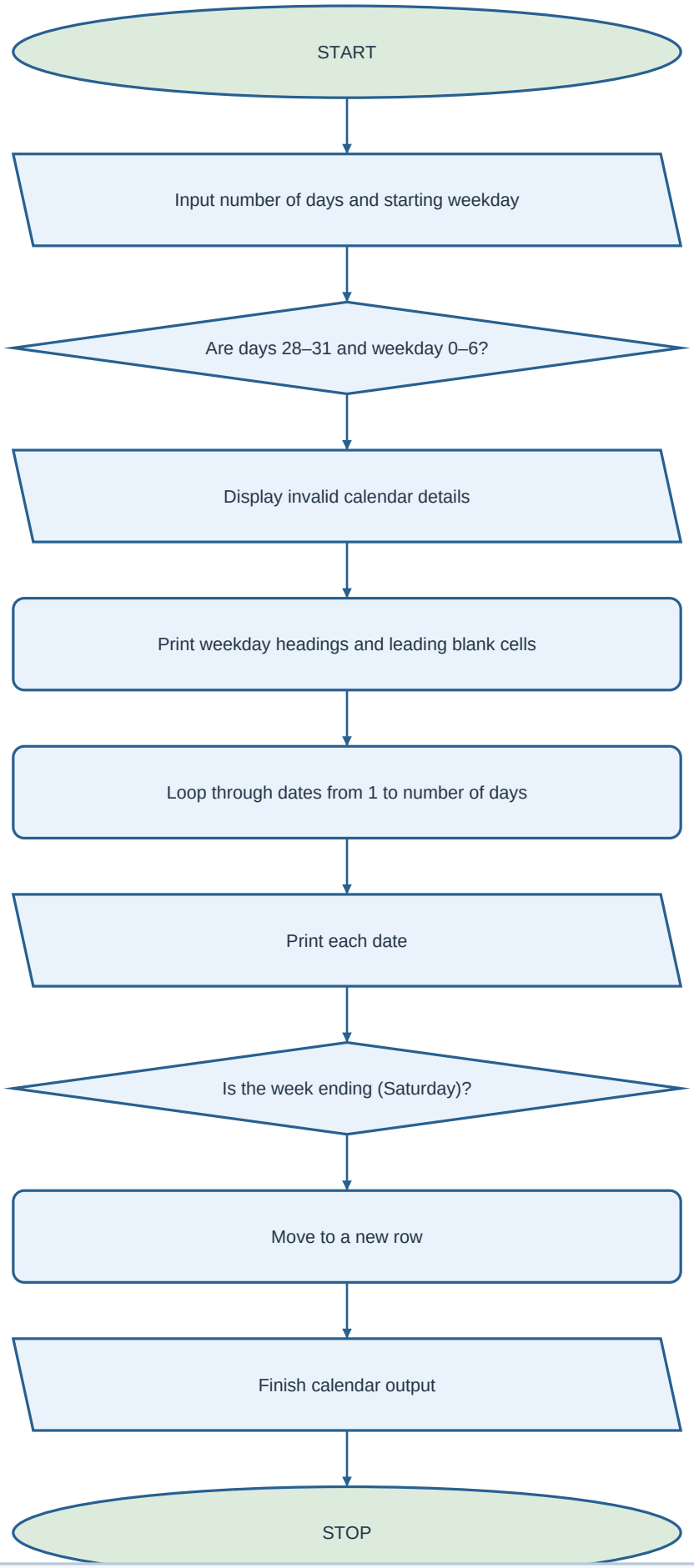
VERTICAL FLOWCHART



Decision paths: YES: Display empty result | NO: Process characters | YES: Increase count | NO: Append run

Case Study 8: Nested Loop Calendar Generator

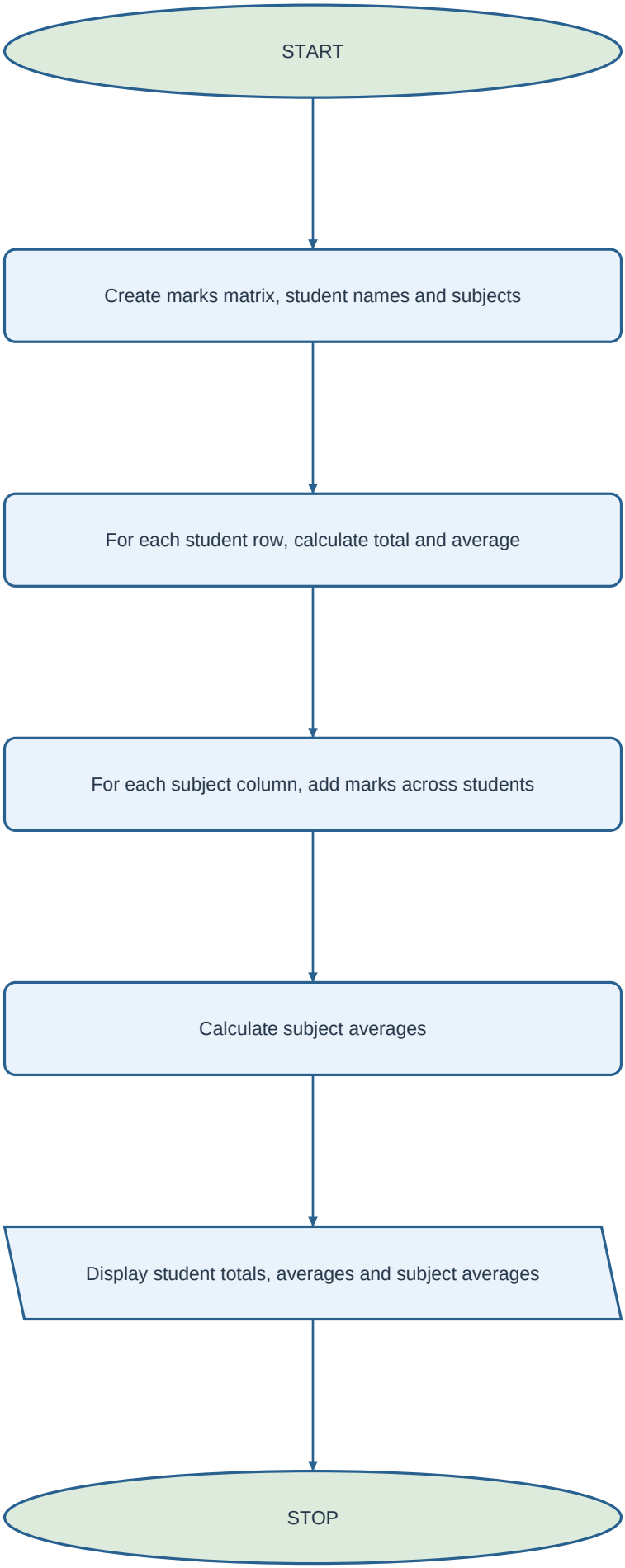
VERTICAL FLOWCHART



Decision paths: NO: Display invalid details | YES: Generate calendar | YES: Start new row | NO: Continue same row

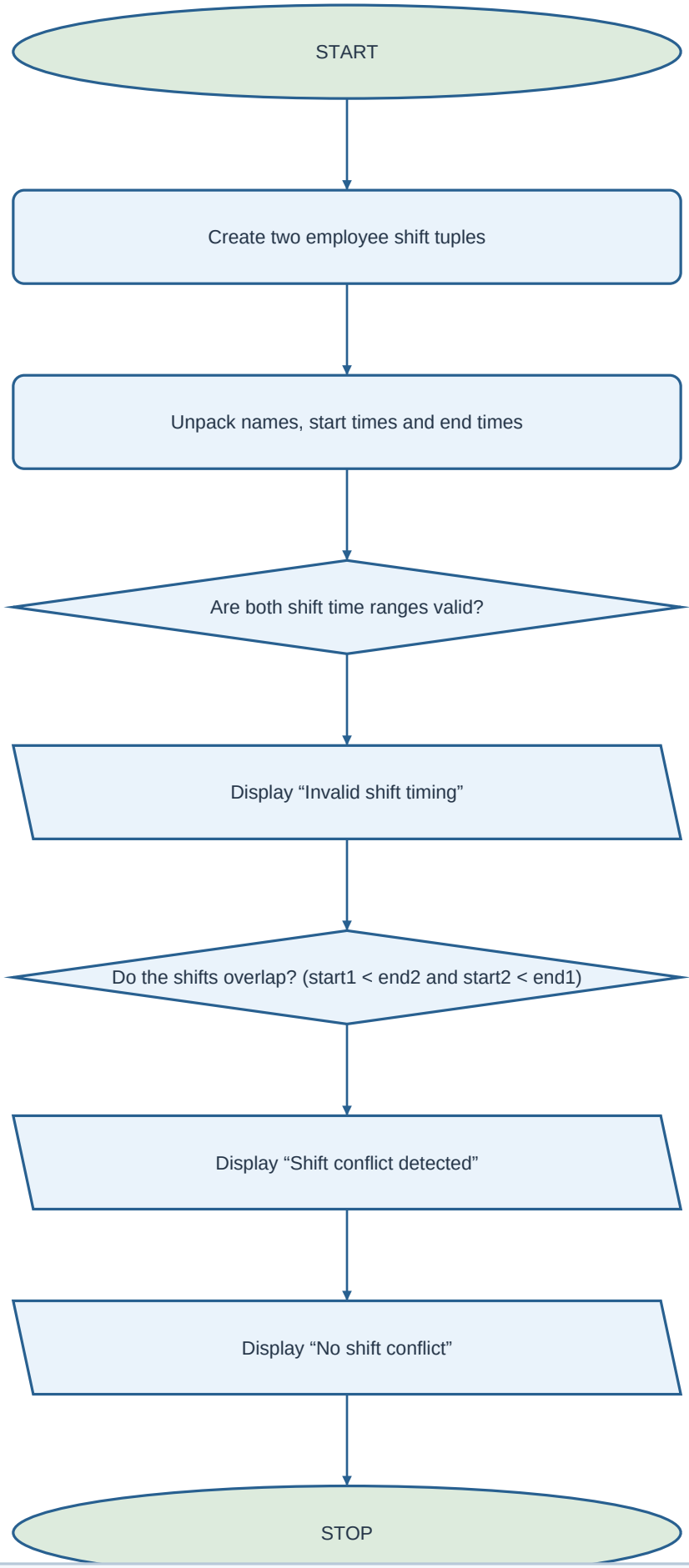
Case Study 9: Student Marks Matrix Analysis

VERTICAL FLOWCHART



Case Study 10: Tuple-Based Employee Shift Conflict Detection

VERTICAL FLOWCHART



Decision paths: NO: Invalid timing | YES: Check overlap | YES: Conflict detected | NO: No conflict